

Clinical characteristics of antidepressant treatment escalation in major depressive disorder in All of Us

1. Introduction

Major depressive disorder (MDD) is the most prevalent mental disorder, affecting over 300 million people in the world¹. Around 21 million Americans experience at least one episode of MDD in 2021; one-third suffer from treatment-resistant depression (TRD), defined as a failure to get a response or remission after at least two trials of antidepressant monotherapy with adequate dose and duration²⁻⁴. Some researchers propose that a patient who does not respond to pharmacotherapy for at least 6 to 8 weeks or to a psychotherapy trial for a minimum of 10 to 18 weeks is classified as a TRD patient⁵. However, a reduction in symptoms of less than 50% is commonly used as a criterion for TRD. While the burden of TRD on the U.S. economy was calculated as 43.8 billion in 2021, the reasons for finding a functional therapeutic approach go beyond the costs, as half of TRD patients report lifetime suicidal ideation^{6,7}.

Due to the complex mechanism and pathogenesis of MDD, the existing therapeutic approaches remain relatively ineffective for more than 60% of patients experiencing complete remission after receiving proper treatment⁸. The TRD treatment strategies include antidepressant dosage optimization, switching medications, and augmentation with other drugs such as second-generation atypical antipsychotics, ketamine/Esketamine, lithium, or thyroid hormone⁹⁻¹³. Non-pharmacological therapies, including psychotherapy, transcranial magnetic stimulation (TMS), electroconvulsive therapy (ECT), and vagus nerve stimulation, are prescribed to patients who do not respond to drug therapy¹⁴⁻¹⁸. However, most of these therapeutic options, whether used alone or in combination, appear either invasive with intolerable side effects or ineffective for the majority of patients with severe depression.

Some researchers argue that, rather than focusing solely on finding treatments for TRD, greater emphasis should be placed on identifying patients with complex MDD before resistance develops. A recent paper published in *Molecular Psychiatry* suggests that stratification of MDD patients into clinically complex subgroups should occur early in the course of illness. Instead of waiting for patients to fail two or more antidepressant trials, clinicians could use a combination of polygenic risk scores (PRS) and clinical symptom profiling to identify high-risk subtypes at the initial visit. This approach may enable earlier use of alternative treatment strategies, such as metabolic or anti-inflammatory interventions, rather than prolonged cycling through standard monoaminergic antidepressants¹⁹.

One of the major challenges in TRD is the lack of standardized and consistently recorded definition in large-scale clinical dataset²⁰. Electronic health records (EHR) often cannot capture direct measure of treatment response, symptom severity, medication adherence, or reason for

treatment change. Given the heterogeneity of TRD definitions, researchers rely on proxy measures, such as antidepressant switching patterns and augmentation strategies, to define treatment resistant depression in observational studies^{19,21}.

The *All of Us* Research Program provides a massive and diverse real-world data including EHR, survey, and demographic information to investigate treatment resistance using antidepressant treatment escalation as a proxy, including medication switching and augmentation strategies.

The primary objective of this study is to characterize differences between MDD patients with antidepressant treatment escalation and those maintained on a single SSRI across multiple domains. Specifically, we examine demographic factors such as age, sex, race, and ethnicity; clinical characteristics including psychiatric and medical comorbidities; and healthcare utilization patterns, such as frequency of psychiatric visits and medication use. In addition, where available, we explore associations with depression severity using Patient Health Questionnaire-9 (PHQ-9) scores and self-reported family history of psychiatric disorders.

This study aims to improve the phenotypic characterization of treatment resistant depression in large-scale EHR data. Furthermore, these findings will provide a foundation for future investigation, including genome-wide association studies (GWAS), Phenome-wide association studies (PheWAS), and precision medicine approach, by establishing a scalable and reproducible framework for identifying patients with complex depression treatment trajectories.

2. Methods

2.1. Data source and cohort definition

This study used data from the All of Us Research Program Controlled Tier Dataset v8. Participants were identified from a predefined cohort of individuals with MDD using EHR data in the *All of Us* Researcher Workbench. The cohort included patients with at least one recorded diagnosis of MDD based on SNOMED and ICD-9/10 codes mapped through the OMOP Common Data Model.

Participants were included if they had available data across key domains including condition occurrence, drug exposure, measurement, survey, and visit occurrence. After data integration and cleaning, the final analytic dataset included 78,251 participants.

2.2. Variables and data sources

Data were extracted from multiple domains of the OMOP Common Data Model:

- **Demographics(person table):** age, sex at birth, race, and ethnicity
- **Clinical conditions(condition_occurrence):** MDD and comorbidities including anxiety, bipolar disorder, opioid use disorder (OUD), and hypertension²²
- **Medications(drug_exposure):** antidepressants, antipsychotics, ketamine/esketamine, and augmentation therapies

- **Clinical measurements(measurement):** body mass index (BMI)²², systolic and diastolic blood pressure (SBP, DBP)²², C-reactive protein (CRP)²³, thyroid-stimulating hormone (TSH)^{24,25}, vitamin D^{26,27}, and PHQ-9 scores
- **Family history(survey):** self-reported family history of depression, anxiety, bipolar disorder, PTSD, and substance use
- **Healthcare utilization(visit_occurrence):** psychiatric-related visits within one year of diagnosis

Measurement values were standardized across units, and biologically implausible values were removed based on predefined thresholds. Repeated measurements within defined time windows were summarized using mean values at the person level.

2.3. Phenotype definition

TRD was defined using treatment escalation patterns derived from EHR medication data. Participants were classified as TRD if they met at least one of the following criteria:

- a) **Antidepressant switching:** at least two switches between antidepressant medications from different classes
- b) **Augmentation therapy:** concurrent use of an antidepressant with an augmentation agent such as an antipsychotic, lithium, or other adjunctive therapy
- c) **Ketamine/esketamine exposure:**
 - Esketamine use was classified as TRD-related
 - Ketamine use was classified as TRD if recorded at least three times to avoid capturing procedural or anesthesia-related use

The TRD date was defined as the earliest date on which any of these criteria were met. Participants not meeting these criteria were classified as non-TRD.

This operational definition reflects real-world clinical practice²⁸ and is commonly used in observational EHR-based studies where direct measures of treatment response are not available. Lack of response to at least two adequate antidepressant treatment is widely used as TRD definition^{29–31}. Moreover, some papers mentioned the need for augmentation therapy as the defining characteristic the TRD population studied³². While esketamine is the only FDA approved medications specifically for TRD patients, racemic ketamine injection is frequently used off-label for TRD³³.

2.4. Time windows and index date definition

For TRD participants, clinical measurements were evaluated within a window of 180 days before to 180 days after the TRD date. For non-TRD participants, an index date was defined as the first

antidepressant exposure, and the same ± 180 -day window was applied. Healthcare utilization was measured within one year after the index date.

2.5. Statistical analysis

Descriptive analyses were conducted to compare baseline characteristics between TRD and non-TRD groups. Categorical variables were summarized using counts and percentages, while continuous variables were summarized using means with standard deviations or medians with interquartile ranges, as appropriate based on distribution.

Group differences were evaluated using chi-squared tests for categorical variables and Wilcoxon rank-sum tests for continuous variables.

To assess associations between clinical measurements and TRD status, distributions were examined using histograms and boxplots, and group comparisons were performed using nonparametric tests due to skewness in several variables.

A multivariable logistic regression model was conducted to identify independent predictors of TRD (yes/no). Covariates included demographic characteristics (age, sex at birth, race, and ethnicity), clinical comorbidities (anxiety, bipolar disorder, opioid use disorder, and hypertension), family history variables derived from survey data, and healthcare utilization measured as the number of psychiatric visits. Due to right-skewness, visit counts were log-transformed using a $\log_{1p}$ transformation prior to modeling. It is worth mentioning that strict filtering of visit counts was not applied, as high values may reflect important clinical situations, such as hospitalizations or more severe disease, particularly in TRD patients. Only patients with more than 365 visits per year were excluded. Results were reported as odds ratios with 95% confidence intervals.

Model performance was evaluated using receiver operating characteristic (ROC) analysis and the area under the receiver operating characteristic curve (AUC). Multicollinearity among predictors was assessed using variance inflation factors (VIF), with standard thresholds applied to identify potential collinearity concerns.

All analyses were conducted using R (version 4.5.0, 2025-04-11) within a Jupyter Notebook environment in the All of Us Research Program Workbench.

3. Results

3.1. Demographics

A total of 78,251 participants were enrolled in the final dataset, including 33,891 individuals classified as non-TRD and 44,360 as TRD. Baseline demographics such as age, sex, race, and ethnicity were summarized in Table 1.

The TRD group had a slightly older age distribution, with a greater proportion of participants aged 41 to 65 years compared with the non-TRD group. Sex distribution was similar across groups, with both groups predominantly female. Race and ethnicity distributions also differed statistically by TRD status, although the overall patterns were broadly similar. Most participants in both groups were White, followed by Black or African American participants. Hispanic or Latino participants represented a smaller proportion of the TRD group compared with the non-TRD group.

Table1. Baseline demographic, clinical, and healthcare utilization characteristics by TRD status			
Characteristic	Non-TRD N = 33,891¹	TRD N = 44,360¹	p-value²
Age group			<0.001
18–25	2,288 (8.1%)	2,274 (5.3%)	
26–40	7,048 (25%)	9,906 (23%)	
41–65	14,785 (53%)	24,244 (57%)	
>65	4,034 (14%)	6,253 (15%)	
Sex at birth			0.003
Female	22,770 (68%)	30,284 (69%)	
Intersex	Suppressed	Suppressed	
Male	10,725 (32%)	13,525 (31%)	
Race			<0.001
American Indian or Alaska Native	593 (2.1%)	488 (1.3%)	
Asian	495 (1.8%)	423 (1.1%)	
Black or African American	5,416 (19%)	7,413 (19%)	
Middle Eastern or North African	145 (0.5%)	153 (0.4%)	
More than one population	1,775 (6.3%)	2,422 (6.3%)	
Native Hawaiian or Other Pacific Islander	38 (0.1%)	27 (<0.1%)	
White	19,635 (70%)	27,466 (72%)	
Ethnicity			<0.001
Hispanic or Latino	5,979 (18%)	5,608 (13%)	
Not Hispanic or Latino	26,919 (82%)	37,333 (87%)	
Anxiety			<0.001
No	10,541 (32%)	7,784 (18%)	
Yes	22,758 (68%)	36,396 (82%)	

Bipolar disorder			<0.001
No	29,741 (89%)	35,546 (80%)	
Yes	3,558 (11%)	8,634 (20%)	
Opioid use disorder			<0.001
No	30,687 (92%)	38,900 (88%)	
Yes	2,612 (7.8%)	5,280 (12%)	
Hypertension			<0.001
No	13,440 (40%)	14,996 (34%)	
Yes	19,859 (60%)	29,184 (66%)	
Family history: Depression			<0.001
No	25,342 (75%)	30,598 (69%)	
Yes	8,549 (25%)	13,762 (31%)	
Family history: Anxiety			<0.001
No	29,973 (88%)	37,740 (85%)	
Yes	3,918 (12%)	6,620 (15%)	
Family history: Substance use			<0.001
No	30,592 (90%)	39,311 (89%)	
Yes	3,299 (9.7%)	5,049 (11%)	
Medical visits in 1 year	2 (4)	7 (11)	<0.001
Number of unique medications	1.10 (0.60)	3.67 (1.83)	<0.001
¹ n (%); Mean (SD)			
² Pearson's Chi-squared test; Wilcoxon rank sum test			

3.2. Clinical Comorbidities and Family History

Clinical comorbidities were more common among TRD participants (Figure1). Anxiety was present in 82% of the TRD group compared with 68% of the non-TRD group. Bipolar disorder was also more frequent among TRD participants, affecting 20% of the TRD group compared with 11% of the non-TRD group. Opioid use disorder and hypertension were also more prevalent in the TRD group. Family history variables showed similar patterns. A family history of depression was reported in 31% of TRD participants compared with 25% of non-TRD participants, and a family history of anxiety was reported in 15% versus 12%, respectively. These differences were

statistically significant across all listed comorbidity and family history variables (Table1 and Figure1).

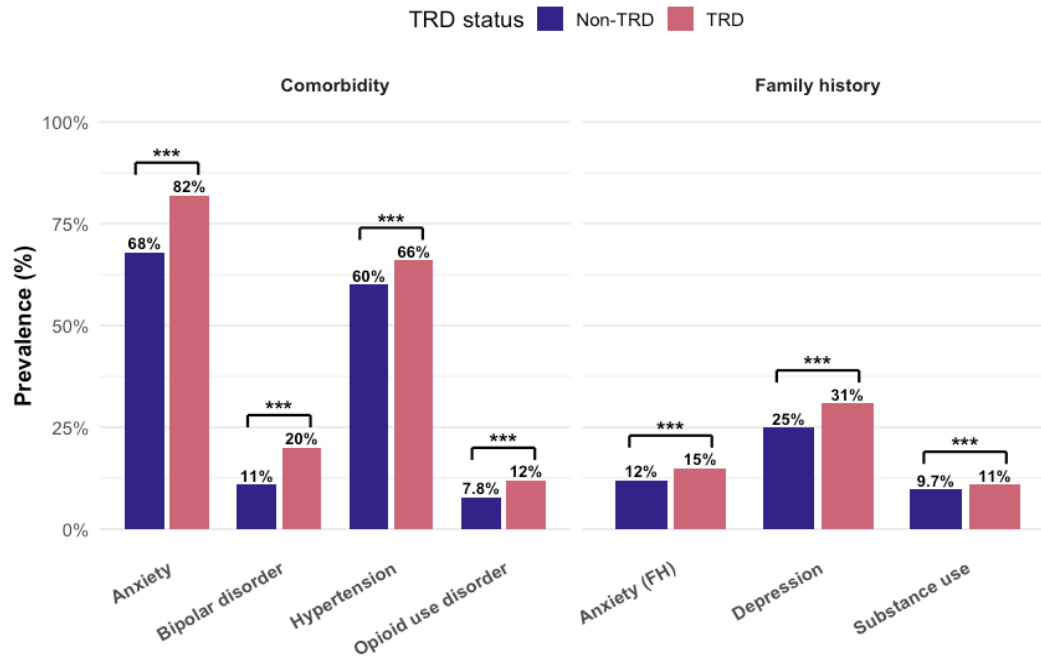


Figure 1. Clinical comorbidities and family history by TRD status. Participants classified as TRD had higher prevalence of anxiety, bipolar disorder, hypertension, opioid use disorder, and family history of anxiety, depression, and substance use compared with non-TRD participants. Asterisks indicate statistically significant group differences based on Pearson's chi-squared tests. *** $p < 0.001$.

3.3. Healthcare utilization and Medication Burden

Healthcare utilization differed significantly by TRD status (Table1). Patients in the TRD group had more medical visits within one year than non-TRD participants. The distribution of visit counts was strongly right-skewed, with most participants having relatively few visits and a smaller subset having very high healthcare utilization. These high-utilization values were retained because they may reflect clinically meaningful patterns, including hospitalization, severe disease, or more intensive clinical management. The TRD group also had a higher number of unique medications, with a mean of 3.67 compared with 1.10 in the non-TRD group, consistent with greater treatment complexity among individuals meeting the TRD definition (Table1).

3.4. Measurements Comparison

Clinical measurement differences between groups were statistically significant for several variables, including BMI, systolic blood pressure, diastolic blood pressure, CRP, TSH, and PHQ-9 score. However, the absolute differences were relatively small. For example, median BMI was 30 in both groups, median systolic blood pressure was 124 in the non-TRD group and 125 in the TRD group, and median PHQ-9 score was 11 versus 12. Vitamin D did not differ significantly

between groups. Overall, these findings suggest that although some clinical measurements were statistically different, they may not meaningfully distinguish TRD from non-TRD participants in this dataset.

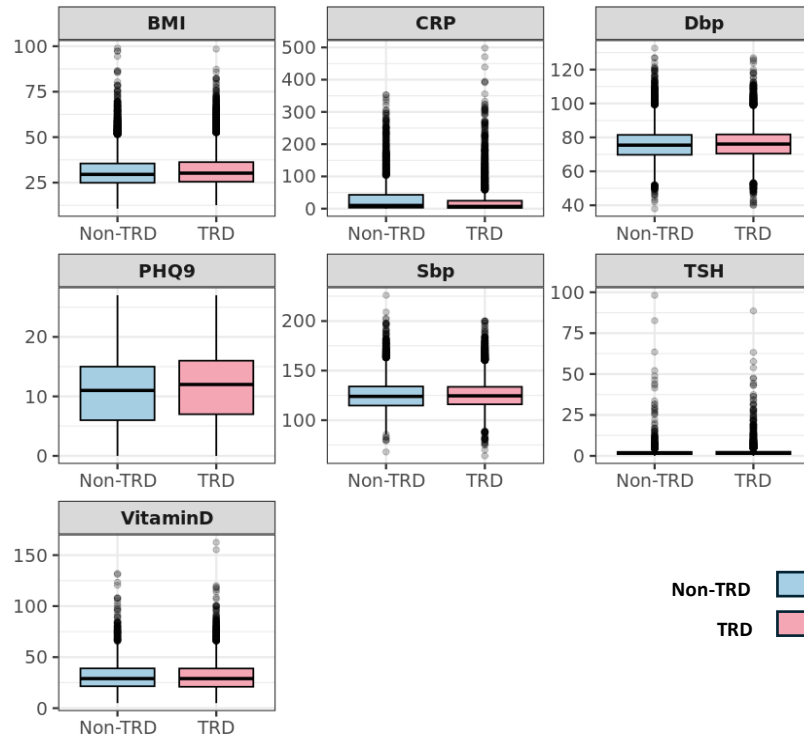


Figure2. Distribution of clinical measurements stratified by TRD status. Although several variables show statistically significant differences between TRD and non-TRD groups, the overall distributions are similar with substantial overlap, indicating that these measurements alone may not strongly distinguish TRD from non-TRD participants. Boxplots represent median, interquartile range, and outliers.

3.5. Logistic Regression Model result

In the multivariable logistic regression analysis, several demographic, clinical, and healthcare utilization variables were significantly associated with TRD status (Table 2). The strongest association was observed for healthcare utilization, as measured by the log-transformed number of medical visits. Patients with higher visit frequency had substantially increased odds of TRD (OR = 2.60, 95% CI: 2.55–2.66, $p < 0.001$).

Psychiatric comorbidities were also strongly associated with TRD. Bipolar disorder demonstrated one of the largest effects, with nearly a two-fold increase in the odds of TRD (OR = 1.94, 95% CI: 1.84–2.05, $p < 0.001$). Similarly, anxiety was significantly associated with higher odds of TRD (OR = 1.80, 95% CI: 1.73–1.88, $p < 0.001$). Hypertension was also significantly associated with TRD, although with a smaller effect size (OR = 1.14, 95% CI: 1.09–1.18, $p < 0.001$).

Demographic factors showed smaller effects. Increasing age was associated with a slight increase in TRD odds (OR = 1.01 per year, $p < 0.001$), while male sex was associated with lower odds compared with females (OR = 0.81, 95% CI: 0.78–0.85, $p < 0.001$). Several race and ethnicity categories were associated with TRD, although these effects were moderate and not consistent across all groups. Overall, the model demonstrated moderate discriminative ability (AUC = 0.766), indicating reasonable performance in distinguishing TRD from non-TRD participants, with no evidence of multicollinearity.

Table2. Multivariable logistic regression results			
Characteristic	OR	95% CI	p-value
Age	1.01	1.01, 1.01	<0.001
Sex (at birth)			
Female	—	—	
Intersex	0.94	0.35, 2.65	>0.9
Male	0.81	0.78, 0.85	<0.001
Race			
American Indian or Alaska Native	—	—	
Asian	0.95	0.77, 1.18	0.7
Black or African American	1.41	1.21, 1.64	<0.001
Middle Eastern or North African	1.09	0.81, 1.48	0.6
More than one population	1.42	1.21, 1.66	<0.001
Native Hawaiian or Other Pacific Islander	1.03	0.55, 1.90	>0.9
White	1.41	1.22, 1.63	<0.001
Ethnicity			
Hispanic or Latino	—	—	
Not Hispanic or Latino	1.28	1.15, 1.42	<0.001
Anxiety	1.80	1.73, 1.88	<0.001
Bipolar disorder	1.94	1.84, 2.05	<0.001
Opioid use disorder	1.46	1.37, 1.56	<0.001
Hypertension	1.14	1.09, 1.18	<0.001
Family history: depression	1.23	1.18, 1.30	<0.001
Family history: anxiety	1.13	1.06, 1.22	<0.001
Family history: bipolar disorder	0.98	0.89, 1.07	0.6
Family history: PTSD	1.23	1.12, 1.35	<0.001
Family history: substance use	0.95	0.88, 1.01	0.12
Visits (Log-transformed)	2.60	2.55, 2.66	<0.001
<i>Abbreviations: CI = Confidence Interval, OR = Odds Ratio</i>			

4. Discussion

This study evaluated demographic, clinical, family history, and healthcare utilization factors associated with TRD using EHR data from the *All of Us* Research Program. Among approximately 78,000 individuals with MDD, more than half met criteria for treatment escalation. Demographic differences between TRD and non-TRD participants were relatively modest, while psychiatric comorbidities, healthcare utilization, and medication use showed much stronger associations with TRD status. Healthcare utilization showed the strongest association in the logistic regression model, suggesting greater clinical complexity and healthcare burden among TRD patients. These findings are generally consistent with prior studies reporting higher psychiatric comorbidity and increased healthcare utilization in TRD patients^{34,35}.

While several clinical measurements, including blood pressure (S_{bp} and D_{pb}), CRP, TSH, and PHQ-9 scores, differed significantly between groups, the magnitude of these differences was relatively small and may not be clinically meaningful. These findings highlight the importance of considering clinical relevance alongside statistical significance, especially in large datasets. Previous studies have suggested associations between complex and non-responsive depression and elevation of inflammatory markers such as CRP³⁶, hypertension³⁷, vitamin D deficiency²⁷, and hypothyroidism^{24,25}. Despite thyroid hormone augmentation being used as a treatment strategy for resistant depression²⁵, TSH values were very similar between groups. Meanwhile, the absence of a clinically meaningful difference in PHQ-9 scores between the TRD and non-TRD groups suggests that treatment resistance may not necessarily reflect greater depression severity. Many patients in the TRD group demonstrated resistance to antidepressant therapy despite having only mild to moderate depressive symptoms.

This study has several limitations. First, there is no universally accepted definition or specific ICD code for TRD, so TRD had to be defined indirectly using treatment escalation patterns such as medication switching and augmentation or administration of ketamine/esketamine. Augmentation therapies may also be prescribed for conditions other than TRD, which could introduce phenotype misclassification. Second, detailed medication information, including dose, duration, adherence, and timing of treatment changes, was not available in the final phenotype definition. Third, healthcare utilization measures were broad and not limited to highly specific psychiatric encounters. Finally, EHR-based studies are subject to missing data, coding inconsistencies, and variability in clinical documentation.

Future studies should focus on improving TRD phenotype definitions using more detailed medication timelines, adherence patterns, and psychiatric-specific healthcare utilization measures. In addition, the availability of genomic data in the *All of Us* Research Program provides an important opportunity for future genome-wide association studies (GWAS), polygenic risk score analyses, and multi-omics investigations aimed at identifying biological mechanisms underlying treatment resistance and improving personalized treatment strategies.

References

1. Trivedi MH. Major Depressive Disorder in Primary Care: Strategies for Identification. *J Clin Psychiatry*. 2020;81(2):UT17042BR1C. doi:10.4088/JCP.UT17042BR1C
2. Michoma D, Couch K. Improving Person-Centered Depression Screening and Care in an Urban Faith-Based Community. *J Dr Nurs Pract*. 2024;17(3):169-176. doi:10.1891/JDNP-2023-0047
3. Vaz GN, Turcato FC, Lima IAV, et al. Genetic ablation of the isoform γ of PI3K decreases antidepressant efficacy of ketamine in male mice. *IBRO Neurosci Rep*. 2024;17:87-95. doi:10.1016/j.ibneur.2024.06.002
4. Li CT. Overview of treatment-resistant depression. *Prog Brain Res*. 2023;278:1-23. doi:10.1016/bs.pbr.2023.03.007
5. Brown S, Rittenbach K, Cheung S, McKean G, MacMaster FP, Clement F. Current and Common Definitions of Treatment-Resistant Depression: Findings from a Systematic Review and Qualitative Interviews. *Can J Psychiatry*. 2019;64(6):380-387. doi:10.1177/0706743719828965
6. Zhdanova M, Pilon D, Ghelerter I, et al. The Prevalence and National Burden of Treatment-Resistant Depression and Major Depressive Disorder in the United States. *J Clin Psychiatry*. 2021;82(2):29169. doi:10.4088/JCP.20m13699
7. Nobile B, Gourguechon-Buot E, Malestroit M, et al. Does depression with current suicidal ideation lead to treatment-resistant depression? Two large naturalistic cohorts of outpatients with depression and current suicidal ideation. *Psychiatry Res*. 2024;342:116249. doi:10.1016/j.psychres.2024.116249
8. Hu L, Wu N, Wang J, Cai D. Network pharmacology combined with experimental verification for exploring the potential mechanism of phellodendrine against depression. *Sci Rep*. 2025;15(1):1958. doi:10.1038/s41598-024-84771-9
9. Pessina E, Martini A, Raffone F, Martiadis V. Cariprazine augmentation in patients with treatment resistant unipolar depression who failed to respond to previous atypical antipsychotic add-on. A case-series. *Front Psychiatry*. 2023;14. doi:10.3389/fpsy.2023.1299368
10. Kajumba MM, Kakooza-Mwesige A, Nakasujja N, Koltai D, Canli T. Treatment-resistant depression: molecular mechanisms and management. *Mol Biomed*. 2024;5(1):43. doi:10.1186/s43556-024-00205-y
11. Tundo A, Filippis R de, Proietti L. Pharmacologic approaches to treatment resistant depression: Evidences and personal experience. *World J Psychiatry*. 2015;5(3):330-341. doi:10.5498/wjp.v5.i3.330

12. Pharmacodynamic Interactions Between Ketamine and Psychiatric Medications Used in the Treatment of Depression: A Systematic Review | International Journal of Neuropsychopharmacology | Oxford Academic. Accessed January 18, 2025. <https://academic.oup.com/ijnp/article/24/10/808/6309481>
13. Prange AJ, Wilson IC, Rabon AM, Lipton MA. Enhancement of imipramine antidepressant activity by thyroid hormone. *Am J Psychiatry*. 1969;126(4):457-469. doi:10.1176/ajp.126.4.457
14. Taylor RW, Coleman JRI, Lawrence AJ, Strawbridge R, Zahn R, Cleare AJ. Predicting clinical outcome to specialist multimodal inpatient treatment in patients with treatment resistant depression. *J Affect Disord*. 2021;291:188-197. doi:10.1016/j.jad.2021.04.074
15. Berlim MT, Van den Eynde F, Daskalakis ZJ. Efficacy and acceptability of high frequency repetitive transcranial magnetic stimulation (rTMS) versus electroconvulsive therapy (ECT) for major depression: a systematic review and meta-analysis of randomized trials. *Depress Anxiety*. 2013;30(7):614-623. doi:10.1002/da.22060
16. Wada M, Nakajima S, Taniguchi K, et al. Effectiveness of sequential bilateral repetitive transcranial stimulation versus bilateral theta burst stimulation for patients with treatment-resistant depression (BEAT-D): A randomized non-inferiority clinical trial. *Brain Stimulat*. 2024;18(1):25-33. doi:10.1016/j.brs.2024.12.1474
17. Galbiati C, Dattilo V, Bortolomasi M, et al. Plasma microRNA Levels After Electroconvulsive Therapy in Treatment-Resistant Depressed Patients. *J ECT*. Published online January 7, 2025. doi:10.1097/YCT.0000000000001100
18. Zanello M, Dibué M, Cornips E, et al. Training and teaching of vagus nerve stimulation surgery: Worldwide survey and future perspectives. *Neurochirurgie*. 2023;69(3):101420. doi:10.1016/j.neuchi.2023.101420
19. Fabbri C. Treatment-resistant depression: role of genetic factors in the perspective of clinical stratification and treatment personalisation. *Mol Psychiatry*. 2025;30(5):2210-2218. doi:10.1038/s41380-025-02899-0
20. McIntyre RS, Alsuwaidan M, Baune BT, et al. Treatment-resistant depression: definition, prevalence, detection, management, and investigational interventions. *World Psychiatry*. 2023;22(3):394-412. doi:10.1002/wps.21120
21. Wang LH, Shih MY, Lin YF, Kuo PH, Feng YCA. Polygenic dissection of treatment-resistant depression with proxy phenotypes in the UK Biobank. *J Affect Disord*. 2025;381:350-359. doi:10.1016/j.jad.2025.04.012
22. Breeden M, Gillis A, Salas J, Scherrer JF. Antidepressant treatment and blood pressure control in patients with comorbid depression and treatment resistant hypertension. *J Psychosom Res*. 2022;153:110692. doi:10.1016/j.jpsychores.2021.110692

23. Osimo EF, Baxter LJ, Lewis G, Jones PB, Khandaker GM. Prevalence of low-grade inflammation in depression: a systematic review and meta-analysis of CRP levels. *Psychol Med.* 2019;49(12):1958-1970. doi:10.1017/S0033291719001454
24. Bauer M, Whybrow PC. Role of thyroid hormone therapy in depressive disorders. *J Endocrinol Invest.* 2021;44(11):2341-2347. doi:10.1007/s40618-021-01600-w
25. Trifu S, Popescu A, Dragoi AM, Trifu AI. THYROID HORMONES AS A THIRD LINE OF AUGMENTATION MEDICATION IN TREATMENT-RESISTANT DEPRESSION. *Acta Endocrinol (Copenh).* 2020;16(2):256-261. doi:10.4183/aeb.2020.256
26. Arathimos R, Ronaldson A, Howe LJ, et al. Vitamin D and the risk of treatment-resistant and atypical depression: A Mendelian randomization study. *Transl Psychiatry.* 2021;11(1):561. doi:10.1038/s41398-021-01674-3
27. Alghamdi S, Alsulami N, Khoja S, Alsufiani H, Tayeb HO, Tarazi FI. Vitamin D Supplementation Ameliorates Severity of Major Depressive Disorder. *J Mol Neurosci.* 2020;70(2):230-235. doi:10.1007/s12031-019-01461-2
28. Treatment-resistant unipolar major depression (major depressive disorder) in adults - UpToDate. Accessed May 5, 2026. <https://www.uptodate.com/contents/treatment-resistant-unipolar-major-depression-major-depressive-disorder-in-adults>
29. Fabbri C, Kasper S, Kautzky A, et al. A polygenic predictor of treatment-resistant depression using whole exome sequencing and genome-wide genotyping. *Transl Psychiatry.* 2020;10:50. doi:10.1038/s41398-020-0738-5
30. Fabbri C, Hagenaars SP, John C, et al. Genetic and clinical characteristics of treatment-resistant depression using primary care records in two UK cohorts. *Mol Psychiatry.* 2021;26(7):3363-3373. doi:10.1038/s41380-021-01062-9
31. Shah SB, Peddada TN, Song C, et al. Exome-wide association study of treatment-resistant depression suggests novel treatment targets. *Sci Rep.* 2023;13:12467. doi:10.1038/s41598-023-38984-z
32. Nuñez NA, Joseph B, Pahwa M, et al. Augmentation strategies for treatment resistant major depression: A systematic review and network meta-analysis. *J Affect Disord.* 2022;302:385-400. doi:10.1016/j.jad.2021.12.134
33. Vekhova KA, Namiot ED, Jonsson J, Schiöth HB. Ketamine and Esketamine in Clinical Trials: FDA-Approved and Emerging Indications, Trial Trends With Putative Mechanistic Explanations. *Clin Pharmacol Ther.* 2024;117(2):374-386. doi:10.1002/cpt.3478
34. Adekanattu P, Olfson M, Susser LC, et al. Comorbidity and healthcare utilization in patients with treatment resistant depression: A large-scale retrospective cohort analysis using electronic health records. *J Affect Disord.* 2023;324:102-113. doi:10.1016/j.jad.2022.12.044

35. Voytenko VL, Street P, VanOrman BT. Psychiatric comorbidities in treatment-resistant depression: Insights from a second-opinion consultation case series. *Psychiatry Res Case Rep.* 2024;3(1):100205. doi:10.1016/j.psycr.2024.100205
36. Cavalcanti-Ribeiro P, De Souza LCAM, de Tavares VDO, et al. Inflammatory response to repeated subcutaneous esketamine in treatment-resistant depression: The role of baseline C-reactive protein. *J Affect Disord.* 2026;397:120877. doi:10.1016/j.jad.2025.120877
37. Inoue T. Depressive symptoms and the development of hypertension. *Hypertens Res.* 2024;47(11):3070-3072. doi:10.1038/s41440-024-01856-8

Supplementary materials are provided as a separate PDF file.

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